A Rail-to-Rail 12 MS/s 91.3 dB SNDR 94.1 dB DR Two-Step SAR ADC With Integrated Input Buffer Using Predictive Level-Shifting

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***Abstract*— Input buffers can be used to reduce the input load of high-resolution discrete-time (DT) Nyquist analog-to-digital converters (ADCs), which can be challenging to drive, particularly at high sampling rates, because of the large input sampling capacitance needed to reduce thermal noise. An input driving technique called predictive level-shifting is proposed to drive rail-to-rail signal swing without increasing the buffer supply rail for high linearity. This article demonstrates an easy-to-drive two-step successive approximation register (SAR) ADC with integrated input buffer, achieving 91.3 dB peak signal-to-noise-and-distortion ratio (SNDR) and 94.1 dB dynamic range (DR) with 6.6 V peak-to-peak differential signal swing and 4 MHz input frequency at 12 MS/s with 30.41 mW power consumption, demonstrating the best performance among the published literature.**

***Index Terms*— Analog-to-digital converter (ADC), correlated double sampling (CDS), input buffer, predictive level-shifting, predictor, ring amplifier, two-step successive approximation register (SAR) ADC.**

I. Introduction

HIGH-RESOLUTION analog-to-digital converters (ADCs) with a dynamic range (DR) greater than 90 dB are essential in medical imaging systems such as digital X-ray to distinguish the distance and composition of tissues through which the X-rays have passed [1]. Power-efficient successive approximation register (SAR) typologies are widely used in such high-resolution ADCs [1], [2], [3], [4], [5]. However, high-resolution discrete-time (DT) Nyquist ADCs are difficult to drive, especially at high sampling frequencies, because of their large input sampling capacitance required to suppress thermal noise.

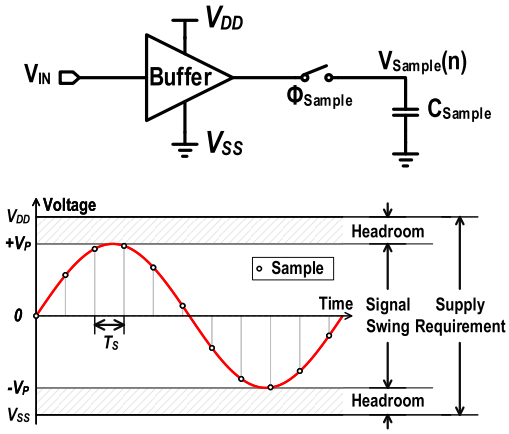


Fig. 1. ADC sampling front-end with conventional input driving method.

Standalone general-use input buffers are costly as a wide supply range is needed to maintain linearity throughout the input range, impairing the power efficiency as shown in Fig. 1. Integrated driving techniques such as [6] have been explored, although the use of the embedded buffer inside the SAR ADC loop requires each SAR decision to resettle through the buffer's bandwidth while also requiring a separate larger power supply (2.5 V) to accommodate the 1.8 Vpp signal swing. ADC input load can also be significantly reduced with the kT/C noise cancellation technique proposed in [7] where the sampling noise is suppressed by a preamplifier gain. However, the sampling window should be small enough to avoid preamplifier linearity degradation due to large signal swing, which in turn requires a high bandwidth preamplifier for desired accuracy [7], [8], [9].

Input buffering with predictive level-shifting is proposed in this article to reduce the input load of the ADC while maintaining high linearity and power efficiency. This technique breaks the input signal into small pieces and moves them into

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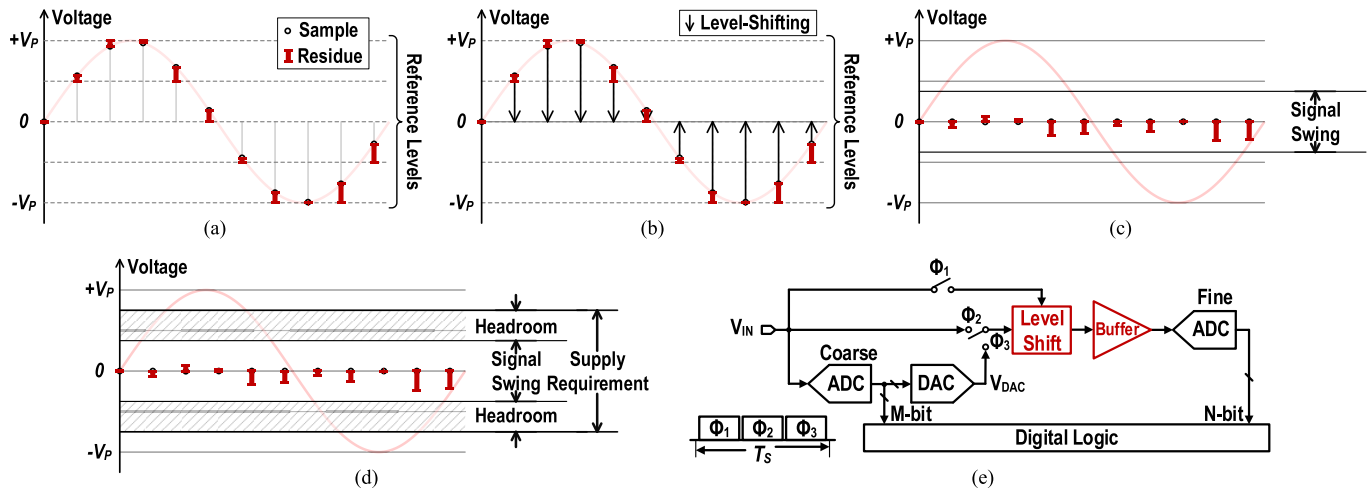


Fig. 2. Signal swing reduction of the input buffer using level-shifting. (a) Samples with coarse quantization levels and residues. (b) Both the samples and residues are level-shifted toward a bias voltage. (c) Buffer signal swing is much smaller than original input signal swing. (d) Supply requirement can be reduced. (e) Block diagram.

a smaller range via analog level-shifting, thus relaxing the swing requirement of the buffer. As a result, rail-to-rail ADC operation is achieved without increasing the buffer supply voltage. To demonstrate the effectiveness of this technique, a prototype two-step SAR ADC with an integrated input buffer is implemented in a 180 nm CMOS technology. The ADC occupies an active area of 0.6 mm^2 and achieves over 94 dB DR for 4 MHz input signal at 12 MS/s sampling speed while consuming 30.41 mW [10].

This article is organized as follows. Section II explains the basic principles of input buffering with predictive level-shifting and explores the implementation of the predictor. Section III provides details of the ADC prototype. The measurement results are presented in Sections IV and V concludes this work.

II. Input Buffering With Predictive Level-Shifting

In a high-accuracy ADC with a heavy input load, the input buffer typically requires a large bias current to achieve a low output impedance for high-frequency input signals. In addition, a wide supply range is necessary to maintain the linearity of the full-swing input signal. These current and supply voltage requirements set the power limitation of the input buffer. However, in a DT ADC, the input signal is sampled at constant intervals and only the signal at the sampling moment is important. This means that if the input signal is sampled after being shifted by a certain voltage and this voltage is recorded by a coarse quantization ADC, the original input signal can be recovered with high fidelity. In this way, the swing of the input signal after being level-shifted can be greatly reduced, which significantly reduces the supply rail requirement of the input buffer and greatly improves the power efficiency.

A. Level-Shifting and Predictive Level-Shifting

Fig. 2 demonstrates how an analog level-shift relaxes the swing requirement of the input buffer for an ADC. As shown in Fig. 2(a), with coarse quantization, the input samples can be separated into two components: a coarse quantization voltage, which corresponds to a reference level, and a residue, which is the difference between the sample and the reference level. In Fig. 2(b), both the input sample and the residue are level-shifted toward a certain bias voltage for input buffer, 0 V for instance. As a result, all samples and residues reside within a much smaller range near the bias voltage compared to Fig. 1, as shown in Fig. 2(c). Therefore, with the headroom for buffer linearity requirement considered, the supply voltage does not need to exceed signal swing, thus making rail-to-rail operation feasible.

Fig. 2(e) shows the block diagram of a buffered ADC system with level-shifting. The coarse ADC generates the M-bit coarse quantization result, and the residue is conserved and passed on through the analog level-shift and the input buffer. The residue is then further quantized by the fine ADC into N bits.

As demonstrated in Fig. 2(b), the voltage and the direction of the analog level-shifting are different in each sampling cycle and depend on the signal being sampled in the cycle. The simplest way to preset the level-shift is directly using the input signal as in Fig. 2(e). However, since sampling happens after the level-shift being set, the sampled signal would swing away from the bias voltage. This swing reduction method is still effective if the signal moves slowly, i.e., if the input signal is at a low frequency. However, issues arise when a full-swing input signal near the Nyquist frequency is required. In this case, the input signal may experience a significant change in magnitude during the sampling phase. For instance, the difference can be as large as 0.71× the full-scale value for a 25% duty-cycle sampling phase. In this case, a predictive level-shift can be introduced by adding a predictor block to preset the level-shift as shown in Fig. 3(a). The predictor generates an advanced, or *predicted*, version of the input signal and presets the level-shift so that the buffer output returns to the bias voltage at the sampling moment. Fig. 3(b) illustrates the typical signal behavior of the signals at the ADC input,

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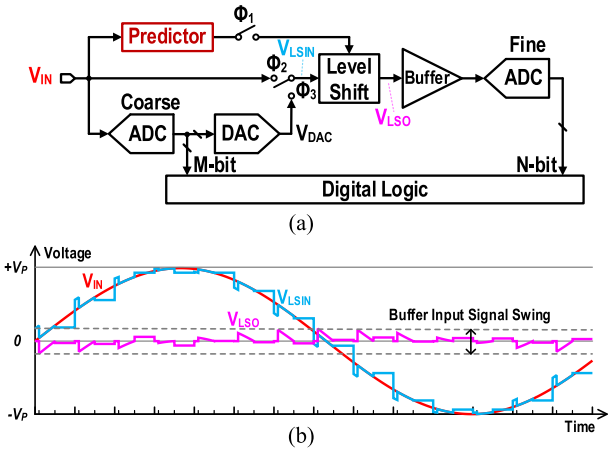


Fig. 3. (a) Predictive level-shifting system and (b) its signal transient behavior.

the level-shift input, and the buffer input, where the effective buffer input signal swing dramatically diminishes compared to the ADC input signal.

B. Operation Principles

The block diagram and operation principle of an ADC applying input buffering with predictive level-shifting are demonstrated in Fig. 4(a)–(e). The system consists of a coarse ADC, a fine ADC, an analog level-shift, a predictor, and an input buffer, as shown in Fig. 4(a). In the example, the capacitor, $C_{\text{Level-Shift}}$, serves as the analog level-shift and C_{Sample} and C_{SAR} comprise the fine ADC capacitor DAC array. The front-end operation consists of three steps: prediction (Φ_{Predict}), sampling (Φ_{Sample}) and residue generation (Φ_{Residue}) as shown in Fig. 4(c)–(e).

In Φ_{Predict}, the predictor generates an advanced version of the input signal (V_{Predict}), connects to the level-shift input node, and sets the voltage difference (ΔV_{LS}) across the level-shift. The other side of the level-shift is reset to the bias voltage (V_{Bias}) of the input buffer. The predictor output leads the input signal by Δt_{P}, which is the same as the duration of the sampling phase. Then in Φ_{Sample}, the level-shift input node switches to the input node. The input signal is level-shifted by ΔV_{LS} and then passed on through the input buffer to the sampling capacitor (C_{Sample}). At the end of the sampling phase, the level-shift input node returns to the sampled V_{Predict}, so the output node also returns to V_{Bias}. Accurate input signal sampling takes place at the end of the sampling phase early edge when the top plate of C_{Sample} is disconnected from the common mode voltage V_{CM}. Coarse quantization also happens at the end of the sampling phase. Then in Φ_{Residue}, the coarse DAC output connects to the level-shift input, so the coarse quantization result is level-shifted by the same ΔV_{LS} and passed on through the input buffer. As a result, the coarse quantization residue is generated at the fine SAR CDAC virtual node. This residue is then further resolved into more bits for higher resolution.

As mentioned above, both the input signal and the coarse quantization result go through the same level-shift offset voltage (ΔV_{LS}) and the same input buffer, therefore, any prediction error, the thermal noise on the level-shift and the flicker noise are canceled due to correlated double sampling (CDS) [7], [11]. Although the prediction error may prevent the buffer input signal from returning to precisely the bias voltage and increases the signal swing the buffer needs to process, this error does not show up in the desired residue and the buffer signal swing is still much smaller than that in the conventional buffering method. In conclusion, the prediction error is absorbed by the linear range of the input buffer, the level-shift does not have kT/C noise requirement and the sampling capacitor is C_{Sample}, which is designed to meet the thermal noise requirement.

C. Prediction

As explained above, a predictor is required to preset the analog level-shift properly. The predictor generates an identical but advanced version of the input signal, and the advance time should be equal to the duration of the sampling phase. Therefore, a predictor should have a unity magnitude response and a constant negative phase delay, or linear positive phase response. The transfer function of an ideal predictor is

$$H(s) = A_0 e^{s\tau} \tag{1}$$

where $A_0 = 1$ for ideal predictor and τ is the desired advancing time. In this prototype, the predictor is designed for 25 ns sampling window with 12 MS/s sampling speed. The time constant of the predictor should be equal to the sampling phase duration, i.e., $\tau = 25$ ns. The prediction should be effective for input signals up to 6 MHz.

An ideal prediction is neither implementable nor expected because of its noncausal property. However, it is possible to approximate the prediction function if the input signal is band-limited. As previously discussed, while the relaxation of input buffer swing requirements may be weakened by practical prediction errors, it can still provide benefits as long as the input signal is moved in the correct direction.

Equation (1) can be approximated with Taylor expansion at $s = 0$ as

$$H(s) = A_0\left(1 + s\tau + \frac{1}{2!}(s\tau)^2 + \frac{1}{3!}(s\tau)^3 + \cdots\right). \tag{2}$$

This is an appropriate approximation when $s \approx 0$, or when the signal is near dc in terms of frequency. These approximations are all high pass filters, and Fig. 5 shows that as input frequency increases, the gain increases and delay decreases. A higher-order approximation yields flatter gain and delay responses that extend to a higher frequency, which translates to smaller gain error, smaller delay error, and larger supported bandwidth. But a higher order of approximation also means more hardware cost for the predictor implementation. There is a trade-off among the accuracy of the prediction, predictor complexity, and signal swing of the input buffer. In cases where a minimized signal swing at the buffer is required, a higher order of predictor can be applied and the coefficients in (2) may be optimized to minimize the prediction error for frequencies of interest.

A first-order approximation of predictor is utilized in this prototype for design simplicity, as shown in Fig. 6. A cheap

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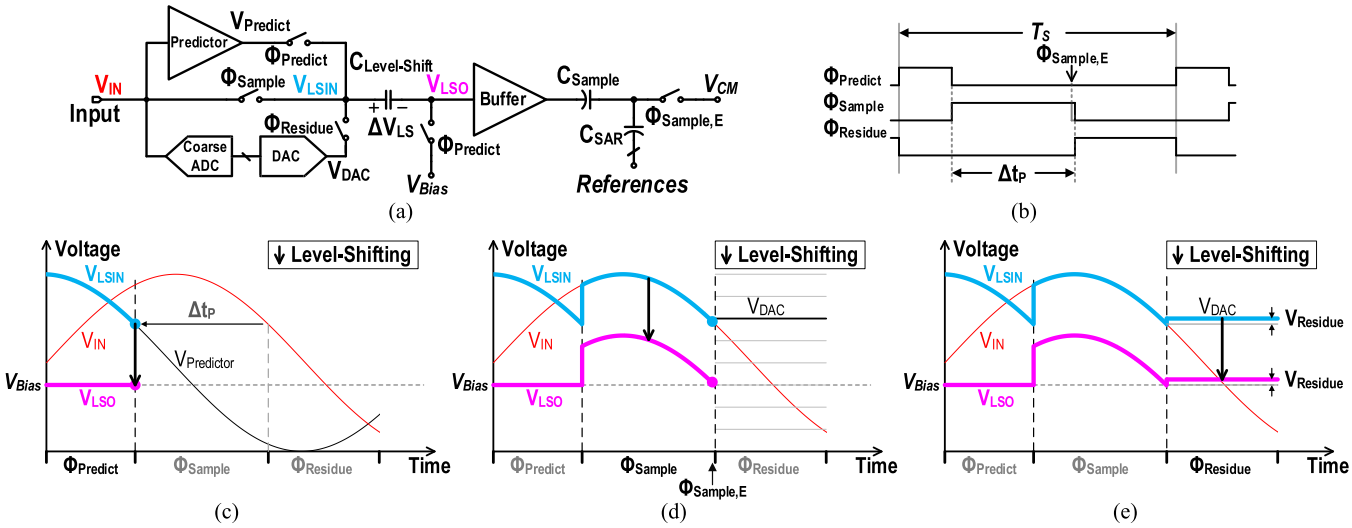


Fig. 4. Input buffering with predictive level-shifting operation concept. (a) System block diagram. (b) Timing diagram. (c) Signal waveform in Φ_{Predict}. (d) Signal waveform Φ_{Sample}. (e) Signal waveform Φ_{Residue}.

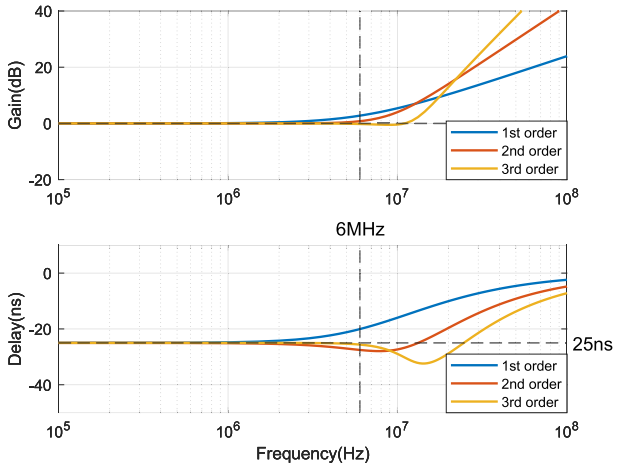


Fig. 5. Gain and delay response of the Taylor expansion approximation of an ideal predictor.

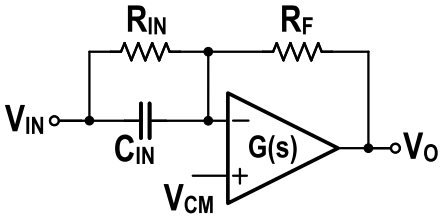


Fig. 6. First order predictor implemented with RC and opamp.

opamp with 16 dB dc gain and 150 MHz bandwidth is used. Therefore, the linearized implemented predictor transfer function of

$$H(s) = \frac{A_0(1+s\tau)}{\left(1+\frac{1}{G(s)}\right)+\frac{A_0}{G(s)}(1+s\tau)} \tag{3}$$

where $A_0 = R_F/R_{\text{IN}}$ is the dc gain, $\tau = R_{\text{IN}}C_{\text{IN}}$ is the time constant and $G(s)$ is the transfer function of the opamp.

Fig. 7 shows the gain responses with different dc gains and delays with different time constants of the first order predictor

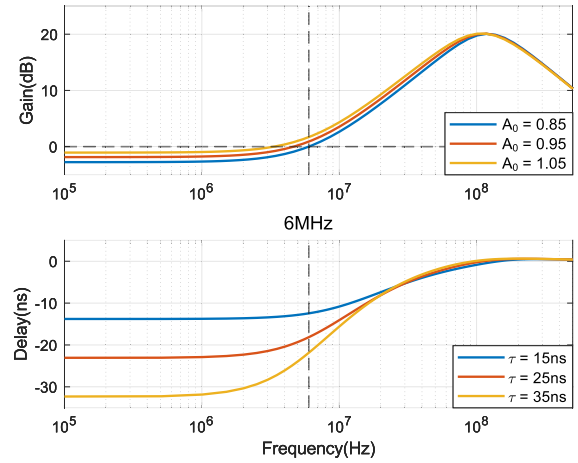


Fig. 7. Gain and delay response for first order predictor with different coefficients: gain response versus dc gain and delay response versus time constant.

using the opamp. Observe that the time constant can be made larger to increase the delay at low frequencies. But for very high frequencies, the delay would be small regardless of the value of the time constant, which means that the circuit is not able to work with a large bandwidth. Since the input signal changes faster in the sampling phase at higher input frequencies and the input buffer usually has a larger linear range for signals at lower frequencies, it is preferable to prioritize the high-frequency accuracy for the predictor. For demonstration, dc gain A_0 is set to be 0.825 V/V so that the gain near 6 MHz is ~0 dB and the time constant τ is set to be 32 ns, which is larger than the sampling window, to optimize the prediction error near 6 MHz without sacrificing too much for lower frequencies.

The buffer input signal swing has two components, prediction error and quantization error. The quantization error is decided by coarse ADC and the prediction error is introduced by nonidealities in prediction. A 4-bit coarse ADC with 3.3-V input range contributes to 0.2-V buffer input signal swing. Smaller prediction errors help reduce buffer signal swing.

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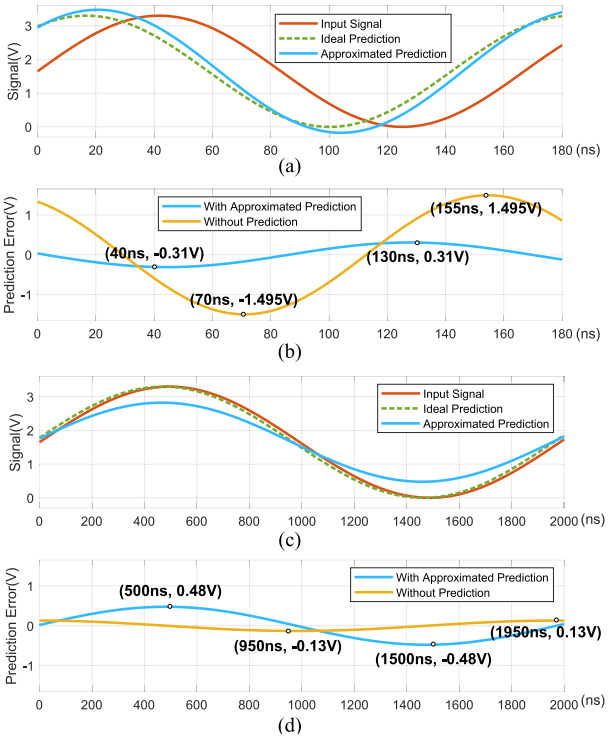


Fig. 8. Transient responses and prediction errors of the predictor with full-swing input signals at different frequencies. 6 MHz for (a) and (b). 0.5 MHz for (c) and (d).

The difference between the predicted signal generated by the implemented predictor and the ideally predicted signal represents the buffer signal swing contribution of the prediction error portion.

Fig. 8 demonstrates the transient responses of the predictor in Fig. 6 and the prediction errors with different input signals. With a 0 dBFS input signal at 6 MHz, the predictor response is compared to the input signal and the ideally predicted signal in Fig. 8(a). The prediction error magnitude, which dominates the signal swing at the buffer input, is ∼0.62 V, compared to ∼2.99 V without the predictor as shown in Fig. 8(b). This is optimized at the cost of larger prediction error at lower frequencies. The predictor response and prediction error for full-swing input signal at 0.5 MHz are shown in Fig. 8(c) and (d), respectively. At lower frequencies, the input signal only changes by a small value during the sampling window. The accommodation of the predictor actually increases the input buffer signal swing, but overall, this is still preferable, as it greatly relaxes the buffer swing requirement near Nyquist frequency. With practical factors taken into consideration, such as circuit saturation for large amplitude input, the prediction error is larger but confined within 1.2-V amplitude for full-swing Nyquist input which needs to be absorbed by the input buffer linear range.

III. Implementation Details

A two-step SAR utilizing the proposed buffering method is implemented. This section presents the implementation details of the prototype, including the system-level details and the circuit details.

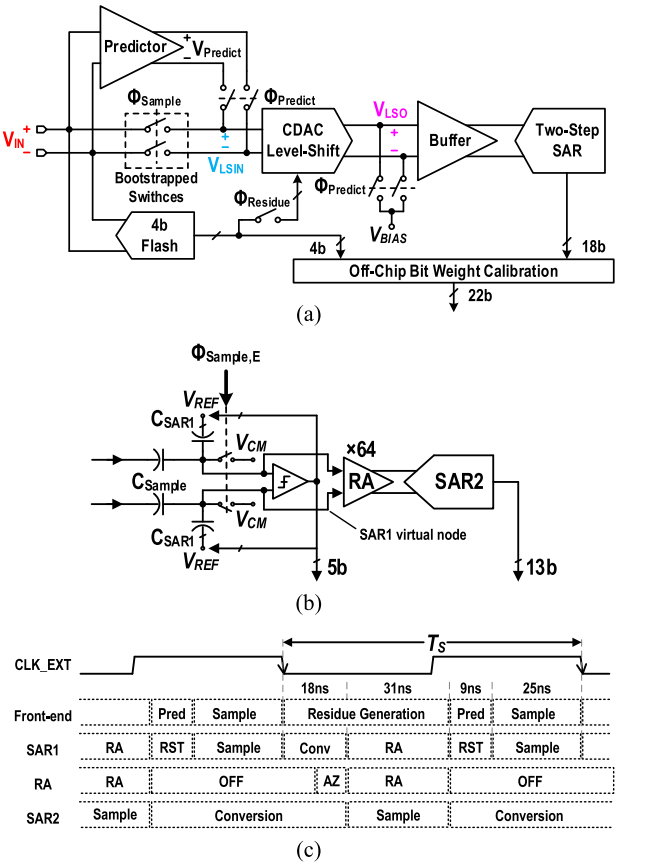


Fig. 9. System details. (a) System block diagram. (b) Fine ADC. (c) Timing diagram.

A. System-Level Details

Fig. 9 shows (a) the system block diagram, (b) the fine ADC implementation, and (c) timing diagram of the prototype. The coarse ADC is implemented as a 4 b Flash ADC and the fine ADC is implemented as an 18 b two-step SAR ADC, resulting in 22 b output digital codes including 6 b redundancy. Foreground calibration is then used to remedy the capacitance mismatch.

The two-step SAR ADC consists of two stages, a 5 b SAR1 and an 18 b SAR2, among which the SAR1 residue is amplified by 64×. After coarse quantization result is generated at the SAR1 virtual node at the beginning of the SAR1 conversion phase, it is resolved by SAR1. Then the SAR1 residue is amplified and further quantized by SAR2. SAR1 sampling capacitance is 17 pF for thermal noise consideration and C_{SAR2} is 2 pF. SAR2 utilizes a 3 + 4 + 6 b bridge structure.

The ADC operates at 12 MS/s sampling speed. In each conversion cycle, the prediction phase takes 9 ns, the sampling phase takes 25 ns and the residue generation phase takes 49 ns. The residue generation phase is further divided into three parts: coarse quantization residue generation, SAR1 conversion, and SAR1 residue amplification. The phase for SAR1 residue amplification is also when SAR2 is sampling the amplified SAR1 residue.

The coarse DAC is merged with the analog level-shift, shown as CDAC Level-Shift in Fig. 10. The bottom plates of the CDAC are switched among predictor output, input signal, and Flash ADC output codes. In the residue generation phase,

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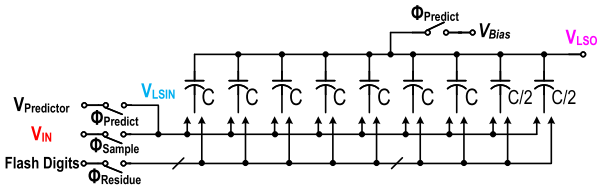


Fig. 10. CDAC level-shift details.

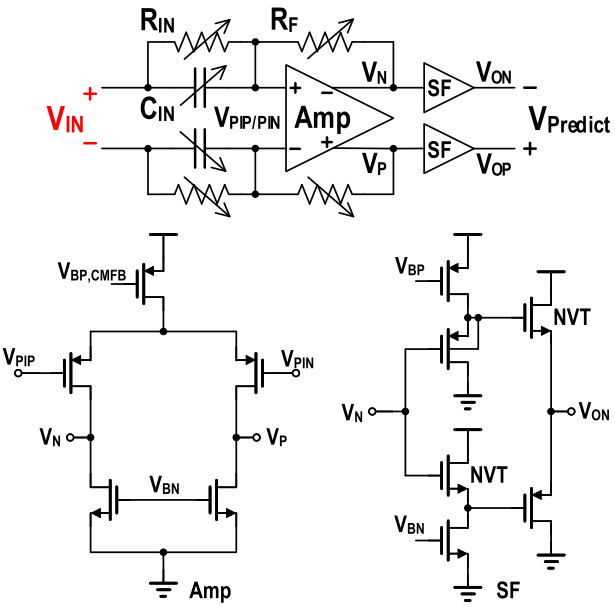


Fig. 11. Predictor implementation.

the CDAC generates an effective coarse quantization result at the bottom of the CDAC.

B. Predictor

The predictor is implemented as in Fig. 11. Tunable resistors and capacitors with nominal values of $R_{\mathrm{IN}} = 40$ kΩ, $R_F = 33$ kΩ and $C_{\mathrm{IN}} = 800$ fF are used. Due to the susceptibility of the resistors and capacitors to PVT variations, their values should be tuned to adjust the dc gain and time constant so that the prediction error amplitude is optimized across the entire bandwidth. Over optimization at one frequency may result in larger prediction errors and deteriorate the performance at other frequencies. The opamp is implemented as a simple five-transistor fully differential amplifier with PMOS differential pair. An additional source-follower stage is used to quickly charge the level-shift capacitors.

The predictor becomes saturated and generates additional undesired prediction errors when the input signal approaches the supply rail. In practice, the prediction error with full-swing input at 6 MHz can be maintained within 1.2 V and is contained within the input buffer linear range to avoid performance degradation. However, additional hardware cost using a higher order predictor with rail-to-rail opamp and buffer may be considered if minimizing prediction error is desired.

C. Input Buffer

The input buffer passes the continuous input signal during the sampling phase and settles the coarse quantization residue

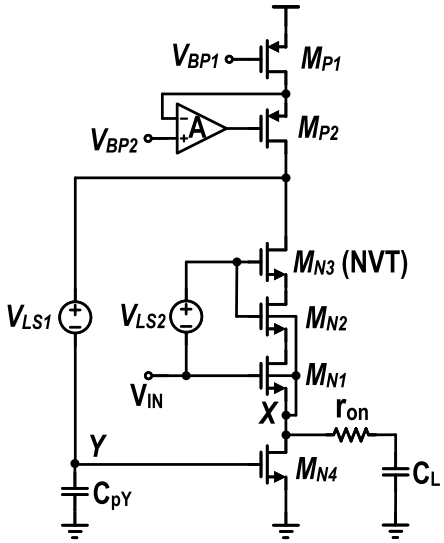


Fig. 12. Flipped-voltage-follower circuit for analysis.

steps during the residue generation phase. These operations impose different requirements on the input buffer, as it must maintain the linearity of the continuous input signal during the sampling phase while accurately settling the coarse quantization residue steps during the residue generation phase. These requirements translate into both linearity and bandwidth requirements for the input buffer. Ideally, with perfect prediction, the swing requirement for the input buffer would only need to tolerate a coarse quantization error with a reasonable margin. However, the prediction error, the amplitude of which can be as large as 1.2 V, plays a more important role. Therefore, in order to tolerate the prediction error and the quantization error with some margin, the input buffer is designed to provide a linear signal swing of 1.5 V for signal frequency up to 6 MHz with over 95-dB linearity. Note that the input buffer does not need to support rail-to-rail swing, but the largest slope of the input signal remains the same as in a conventional topology, or slightly smaller considering the attenuation introduced by the parasitic capacitance at the CDAC level-shift top plates.

A flipped-voltage-follower structure [12] is applied in this design, as shown in Fig. 12. High linearity is achieved by maintaining low output impedance at node X for high bandwidth and high output impedance at node Y for small bias current variation. The transconductance of M_{N1} determines the wide bandwidth and low output impedance for the buffer during the sampling phase and the residue settling phase. The bootstrapped voltage on the gates of M_{N2} and M_{N3} reduces the drain-source voltage variation of M_{N1} and boosts the NMOS output impedance at node Y. The output impedance of the current source M_{P1} is enhanced by gain-boosted cascode device M_{P2} and a common-source gain A. High transconductance in M_{N4} suppresses the voltage variation at node Y resulting from the signal-dependent continuous output current during the sampling phase and impulse settling current during the residue generation phase.

This flipped-voltage-follower has a dominant pole at node Y, $p_1 = 1/r_{oY}C_{oY}$, a nondominant pole at node X, $p_2 = g_{mn1}/(1 + r_{\mathrm{on}}g_{mn1})C_L$, and a zero $z_1 = 1/r_{\mathrm{on}}C_L$,

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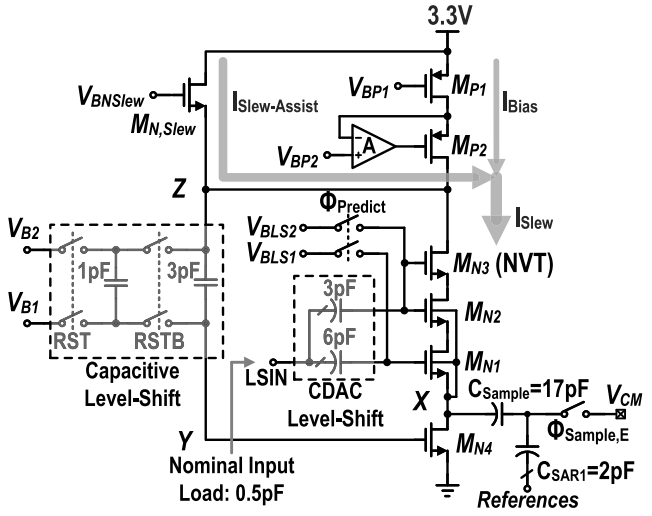


Fig. 13. Input buffer implementation details.

where g_{mnx} is the transconductance of M_{Nx}, C_{pY} is the parasitic capacitance at node Y, C_L is the load capacitance, r_{oY} is the output resistance at node Y and r_{on} is the switched-on resistance of the sampling switch. The flipped-voltage-follower has a local loop gain of

$$\text{LG}(s) \approx -\frac{g_{mn4}r_{oY}}{1+sr_{oY}C_{pY}}\frac{1+sr_{\text{on}}C_L}{1+s\left(r_{\text{on}}+\frac{1}{gmn1}\right)C_L}. \quad (4)$$

The g_{mn1} requirement for stability is relaxed by the damping provided by the zero due to r_{on}. In this design, g_{mn1} has a nominal value of 31 mS and 38 mS for g_{mn4}.

The input buffer obtained is shown in Fig. 13. In this flipped-voltage-follower, the major noise contributors are M_{N1} and M_{P1}. M_{P1} is designed for smaller transconductance and M_{N1} is designed for high transconductance to suppress their noise contribution. An NMOS slew-assist device, M_{NSlew}, provides extra supply-side current to support large signal slewing current, which is otherwise limited by the low-transconductance PMOS current source. The gate voltage of M_{NSlew} is set so that it is cut off during steady-state. In addition, since the buffer has a bandwidth much higher than that of the residue amplifier, the noise can be filtered and further suppressed during the residue amplification phase.

As the buffer's input device is scaled for high transconductance, a larger capacitance for CDAC Level-Shift is used to mitigate the distortion effects of the nonlinear parasitic capacitance of the input device. A secondary CDAC level-shift path is introduced to accommodate the triple cascode bootstrap structure which reduces the drain-source voltage variance of the primary input device (M_{N1}) for better linearity. In total, 6 pF capacitance is used in the primary CDAC Level-Shift path while 3 pF capacitance is used in the secondary CDAC Level-Shift path, and both are implemented as shown in Fig. 10. Native threshold-voltage (NVT) NMOS is used along with a standard threshold device to form the top cascode devices (M_{N2} and M_{N3}). The gates are reset to appropriate biasing voltages during the prediction phase while the input side of CDAC level-shift (LSIN) is connected to the predictor output. A set of capacitive level-shift with a total capacitance of 4 pF is utilized to maintain a constant voltage difference of ~1.9 V ($V_{B2} - V_{B1}$) between nodes Y and Z so that the buffer swing can be optimized.

According to simulation results, the total input capacitive load of the proposed ADC is set by the input CDAC Level-Shift connected in series with the input capacitance of the buffer. This input load appears as 0.5 pF and is only 3% of the sampling capacitance ($C_{\text{Sample}} = 17$ pF).

D. Residue Amplifier

Ring amplifier has been proven to be suitable for residue amplification in two-step SAR ADCs and pipelined ADCs [3], [13], [14], [15]. The dual-deadzone ring amplifier topology is proposed in [3], which uses different devices in separate paths of the output stage to control two main functions of the ring amplifier: 1) large-width HVT devices in deadzone-1 (DDZ1) branch for high impulse current and fast initial charging of load capacitor and 2) cascoded LVT devices in deadzone-2 (DDZ2) branch for low quiescent current, high dc gain and low steady-state bandwidth for kT/C noise filtering. However, since HVT and LVT devices in the third stage share the same bias voltage, the design freedom is constrained in a CMOS process without different threshold voltages. In addition, due to the Miller effect, the parasitic gate-drain capacitance (C_{P}) of the large HVT device is amplified by the third stage gain and thus greatly attenuates the effective signal applied at the gates (V_{G}) and thus results in a significant gain reduction. In [3] this issue is resolved by inserting a pair of neutralization dummy capacitors (C_{Ntr}) between the differential pairs.

In this work, different sets of deadzone capacitors are used in the dual-deadzone paths. This allows the two branches to be biased differently so they can use devices with the same threshold voltage while preserving the benefits of fasting-charging and high-gain in a conventional dual-deadzone ring amplifier. This implementation also eliminates the need for neutralization capacitors due to the absence of the large Miller capacitance in the high gain path (V_{G2}), as shown in Fig. 14.

IV. Measurement Results

The ADC prototype is fabricated in a 180 nm CMOS technology with a die size of 2 × 1.3 mm and occupies an active area of 0.6 mm^2, as shown in Fig. 15. Fig. 16 illustrates the measurement setup for the ADC. With an integrated input buffer, no active components are inserted between the signal generator and the ADC. The test sine wave is produced by the signal generator. Its harmonics and out-of-band noise are suppressed by the passive bandpass filter. The balun doubles the input signal amplitude so that it can be close to the full scale. The two 100 Ω resistors are used to match the signal generator output impedance, and the 140 pF capacitor is used to further suppress high-frequency noise.

The output spectra are shown in Fig. 17. With 2 and 4 MHz with ~−1 dBFS input signal, the ADC achieves over 100 dB SFDR and over 91 dB signal-to-noise-and-distortion ratio (SNDR). As shown in Fig. 18, a DR of 94.1 dB is

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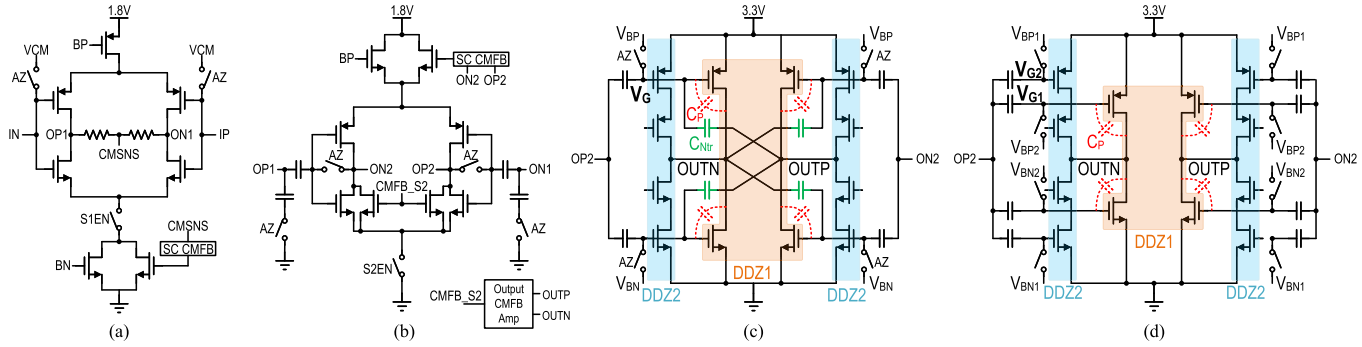


Fig. 14. Ring amplifier details. (a) First stage. (b) Second stage. (c) Output stage in [3]. (d) Proposed output stage.

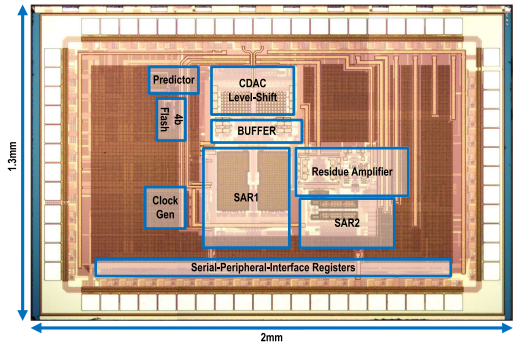


Fig. 15. Die photo.

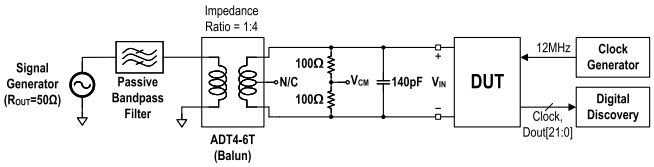


Fig. 16. Measurement setup.

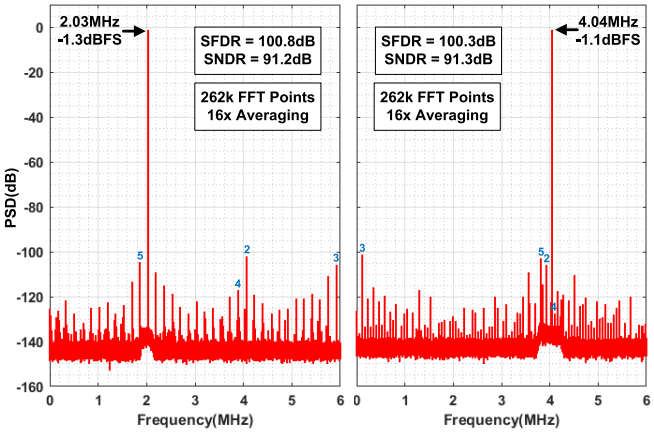


Fig. 17. Measured ADC output spectra at Fs = 12 MS/s with input frequencies of 2.03 and 4.04 MHz.

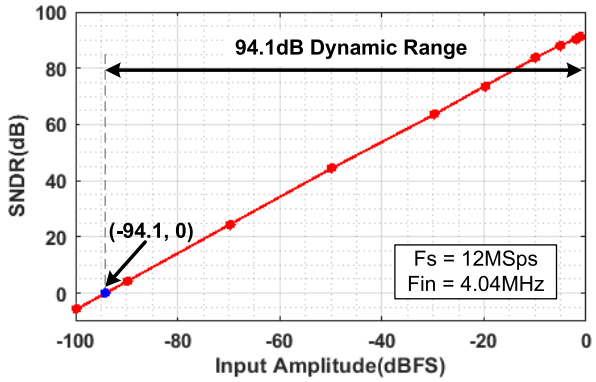


Fig. 18. Measured DR with 4-MHz input frequency at 12 MS/s.

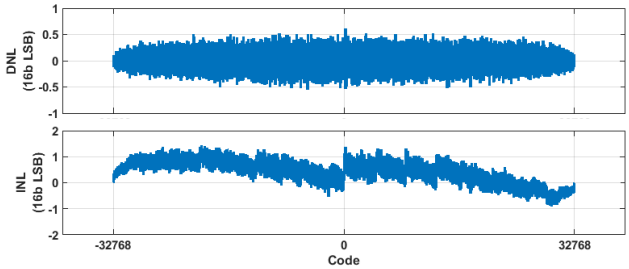


Fig. 19. Static nonlinearity based on 16 b resolution.

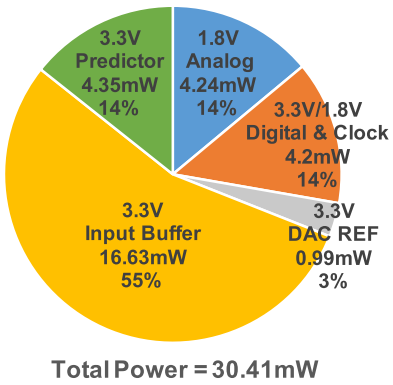


Fig. 20. Power breakdown at 12 MS/s.

achieved. The static nonlinearity based on 16 b resolution is shown in Fig. 19.

The ADC operates at 12 MS/s with 1.8 V/3.3 V supply and 3.3 V reference. The predictor is supplied with 3.3-V power supply while consuming 4.35 mW, as shown in Fig. 20. The input buffer and the third stage of the ring amplifier also have a supply voltage of 3.3 V with 16.63-mW power consumption. The supply of 1.8 V powers the rest of the analog hardware with 4.24-mW dissipation, dominated by the first two stages of the residue amplifier. 4.2-mW digital power with 1.8-V supply

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	This Work	Steensgaard [2] ISSCC 22	ElShater [3] ISSCC 19	Li [4] ISSCC 18	Hummerston [5] VLSI 17	Ho [16] ISSCC 15
Architecture	**Pipelined SAR**	SAR	Pipelined SAR	Pipelined SAR	Pipelined SAR	CTDSM
Technology [nm]	**180**	180	180	180/500	180	55
Area [mm^2]	**0.6**	0.783	1.82	4	3.87	0.09
Speed [MS/s]	**12**	2	15	1	2	140 (BW = 2.2)
SNDR [dB] @ Fin [MHz]	**91.3 @ 4.04**	106 @ 0.001 105.3 @ 0.1	90.8 @ 0.025 88 @ 4	101.5 @ 0.001	99.2† @ 0.1	90.4 @ 0.625
SFDR(THD) [dB] @ Fin [MHz]	**99.2 @ 4.04**	-133 @ 0.001 -114 @ 0.1	100.7 @ 0.025 93dB @ 4	132.5 @ 0.001	-107.0 @ 0.1	104 @ 0.625
DR [dB]	**94.1**	106.3	93.9	102.7	100.5	92
Power [mW]	**9.43*/30.41**	8.5	9.82	19.7	11.35	4.5
FoM$_{S,SNDR}$ [dB]	**179.3*/174.2**	186.0	179.6	175.5	178.6	177.3
FoM$_{S,DR}$ [dB]	**182.1*/177.0**	187.0	182.7	176.7	179.9	178.9
FoM$_W$ [fJ/step]	**26.2*/84.4**	28.2	31.8	202.8	76.1	37.8

FoM$_{S,SNDR}$=SNDR+10log(BW/Power) FoM$_{S,DR}$=DR+10log(BW/Power) FoM$_W$=Power/(2xBWx2$^{(SNDR-1.76)/6.02}$)
* Not including predictor and input buffer.
† Information reported is calculated based on other performance numbers provided.

Fig. 21. Performance summary of the ADC and comparison to the state-of-the-art.

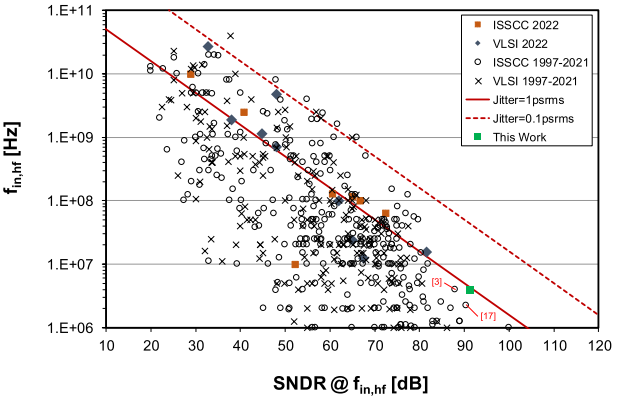


Fig. 22. Input frequency versus SNDR comparison from data in [17].

is distributed among the digital logic, DAC switch drivers, and clock generator. DAC consumes 0.99-mW reference power.

A state-of-the-art comparison is made as shown in Figs. 21 and 22. This is the only work that includes an integrated buffer solution and the power overhead needed to drive the ADC with measurements based on 4 MHz input frequency. This ADC demonstrates the highest performance within published literature of 85+ dB SNDR for >1 MHz input, achieving a 3.3 dB SNDR improvement (4 MHz input frequency) when compared with [3] and a 2.73× speed (BW) improvement when compared with [16].

V. Conclusion

In this work, a technique to enable high-precision signal acquisition of rail-to-rail Nyquist input signals is presented. By utilizing predictive level-shifting, 4-MHz input signal is digitized with an SNDR of 91.3 dB without the use of extended supply-range buffers. The integration of the input buffer also reduces the input loading of the DT converter to 3% of the sampling capacitance enabling conversion without the use of external active drivers. As a result of this technique, the state-of-the-art is improved by 3.3-dB SNDR in accuracy and 2.73× in speed.

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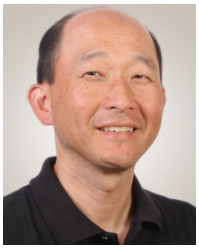
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